

Sales & Customer Retention Analysis

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Insights Derived from SQLite Multi-Table Analysis

Executive Summary

This analysis evaluates sales performance, customer loyalty, and operational efficiency across 7 interconnected data tables.

- **Retention Strength:** Identified a high-loyalty customer base with an 86% repeat purchase rate.
- **Operational Focus:** Preliminary data suggests a need to optimize the "Stock-to-Order" window to minimize return risks.
- **Regional Strategy:** Sales distribution shows varying engagement levels across key global territories.

Methodology & Data Cleaning

Before analysis, a rigorous data cleaning process was executed to ensure technical reliability:

Null & Invalid Key Handling:

Validated all foreign keys; removed "botched" customer entries to prevent downstream errors in relationship mapping.

Format Standardization:

Converted text-based date strings into standardized SQLite-compatible formats for time-series calculations.

Deduplication:

Verified uniqueness across OrderNumbers to ensure volume metrics were not artificially inflated by redundant entries.

User Behavior: The Retention Leaderboard

Distribution of one-time versus repeat users across the lifetime of the dataset.

15,089

REPEAT USERS

2,327

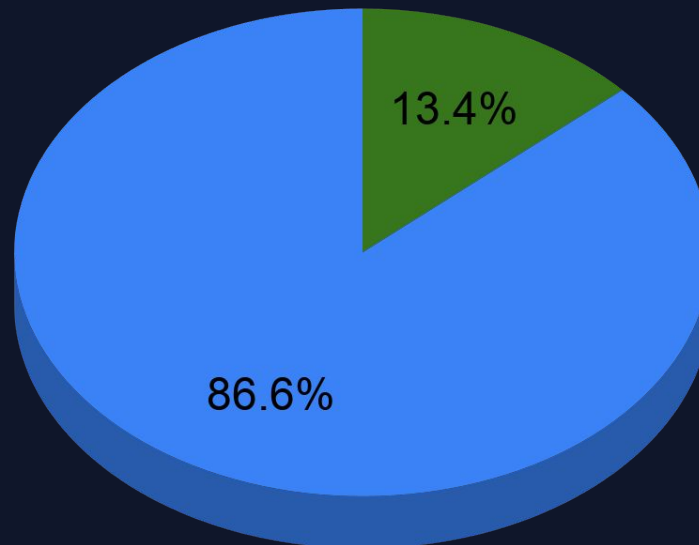
ONE-TIME USERS

86.6%

RETENTION RATE

Our business model is driven by extreme brand loyalty. 86.6% of our customers are repeat buyers, which significantly reduces our long-term customer acquisition costs.

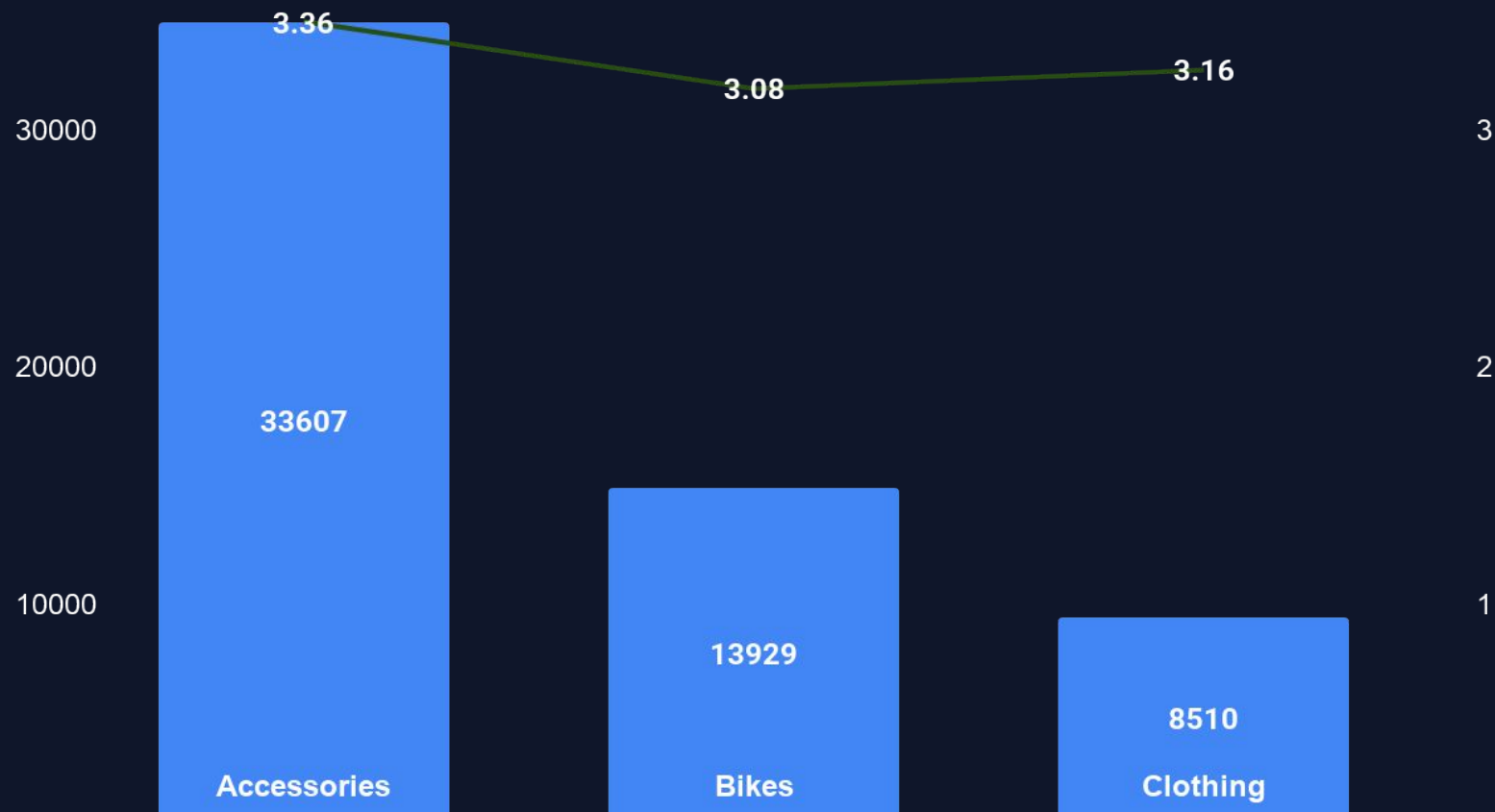
Recommendation: Focus on a 'Loyalty Tier' program to reward the 15,000+ repeat users and ensure they don't churn to competitors.



Product sales

Key Insight: Accessories is our powerhouse, driving 33,607 orders—more than Bikes and Clothing combined. Despite the high volume, the return rate remains stable at 3.36%.

Order Volume vs. Return Rate by Product Category



Regional Activity Distribution

How do user types vary across different territories and continents?

	Continent	Country	Region	total_orders	repeat_order_count	repeat_rate_pct	total_returns_qty	return_rate
1	Pacific	Australia	Australia	12409	205	1.65	404	3.26
2	North America	United States	Southwest	11463	14	0.12	362	3.16
3	North America	United States	Northwest	8267	14	0.17	270	3.27
4	North America	Canada	Canada	6875	798	11.61	238	3.46
5	Europe	United Kingdom	United Kingdom	6423	81	1.26	204	3.18
6	Europe	Germany	Germany	5289	53	1.0	163	3.08
7	Europe	France	France	5239	102	1.95	186	3.55
8	North America	United States	Southeast	34	0	0.0	1	2.94
9	North America	United States	Northeast	27	0	0.0	0	0.0
10	North America	United States	Central	20	0	0.0	0	0.0

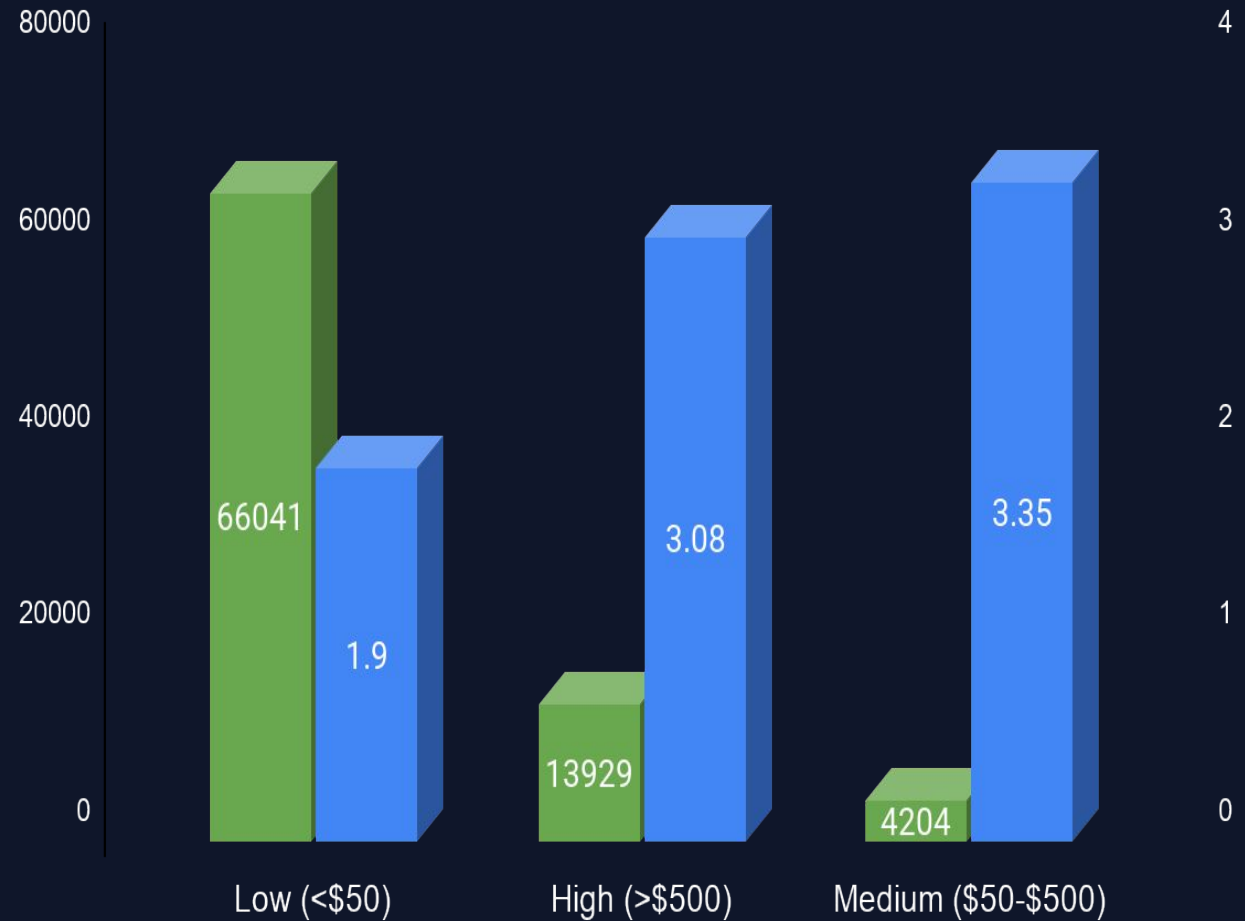
Observation: While Australia has the most orders (12,409), Canada has a massive 11.61% repeat purchase rate. Most other regions are below 2%.

| The Price Effect on Sales

While Medium-tier items (\$50–\$500) have the highest return rate (3.35%), our Low-tier items have a remarkably low return rate of 1.9%

Recommendation:

Investigate the Medium-tier products. A 3.35% return rate on items up to \$500 represents a higher dollar-value loss than returns on \$20 accessories.



Strategic Recommendations

1. Profit Margin Protection:

Conduct a quality audit on the "Medium-Tier" price category. Since these items have our highest return rate (3.35%)

2. New User Conversion:

By successfully converting just 500 of these 2,327 one-time users, we can increase our monthly active customer base by nearly 4%."

3. Inventory Refinement:

There is a significant inventory gap in the Components category. Despite having 14 sub-categories, there were zero retail sales recorded.

Recommendation: Verify if these are internal-only parts (for the repair shop) or if there is a data-tracking error or listing issue preventing customers from buying these parts.

**Thank
You**